INTERNSHIP REPORT

Design and Development of a Full-Stack Web Platform for an Architecture and Interior Design Firm

Six-Month Internship Report on the Full-Stack Developer & Design Intern role undertaken at Limitless Spacez, Greater Noida

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Submitted By

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Institution

B.Tech Computer Science and Engineering

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PROJECT DESCRIPTION

Internship Overview

Organization	Limitless Spacez, Greater Noida, Uttar Pradesh
Role	Full-Stack Developer & Design Intern — Full-time, Hybrid
Duration	05 January 2026 – 04 July 2026 (6 months)
Supervisor	Ar. Sanjay Gautam, Founder & Principal Architect
Deliverable	The firm's live public platform at limitlessspacez.com, and the internal tooling to keep it current without a developer present

PROJECT DESCRIPTION

About Limitless Spacez

A premium architecture, interior design and construction firm operating out of Greater Noida, Uttar Pradesh, under the line “Design · Build · Deliver.” The firm is led by Ar. Sanjay Gautam, Founder & Principal Architect, registered with the Council of Architecture (CA/2021/140523), and takes a project from initial design through statutory approval to completed construction.

Architecture

Residential, commercial and institutional design from concept through working drawings

Interior

Interior design, material and furniture specification, and turnkey fit-out

Map Approval

Preparation and submission of building plans for sanction by the local authority

Construction

Execution and site supervision through to handover

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The Problem

A design practice that cannot show its work convincingly, cannot be found by the clients searching for it, and cannot update its own website, loses enquiries it has already earned.

Presentation

Photographic work presented as small images inside blocks of text — the site described the firm, it did not demonstrate it

Enquiry Capture

Enquiries arrived by DM and phone call, with nothing recorded — a lead that is forgotten is indistinguishable from one that never arrived

Maintainability

Every content change required editing markup, so completed projects accumulated on a phone and never reached the site

Discoverability & Performance

Large unoptimised images and minimal metadata made the practice hard to find and slow to load on mobile

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Project Objectives

01 Map the four service verticals into an information architecture a first-time visitor can navigate unaided

02 Design a quiet, photography-led visual system in Figma, documented as reusable tokens and components

03 Implement a responsive front end in HTML5, CSS3 and Tailwind, functional from a 360px viewport upward

04 Model services, projects and images in a normalised MySQL schema behind a Node.js and Express application

05 Build a validated enquiry pipeline that persists every submission and notifies the firm by email

06 Give non-technical staff an administration panel to manage projects, images and enquiries unaided

07 Optimise image delivery and measure Core Web Vitals improvement on a throttled mobile profile

08 Implement technical SEO and WCAG 2.1 Level AA accessibility, and hand over with written documentation

PROJECT DESCRIPTION

Scope of the Project

Included

- Complete Figma design system and page templates at three breakpoints
- Public front end: home, four service pages, project archive, project detail, about, contact
- Express + MySQL server and data layer with migrations and seed data
- Validated enquiry pipeline with spam mitigation and email notification
- Authenticated administration panel for staff
- Performance, SEO and accessibility work
- Production deployment and written handover documentation

Excluded

- Online payments and any financial transaction handling
- A client-facing project tracking portal (deferred to future scope)
- Native mobile applications
- Architectural drawing, BIM/CAD, or professional design services
- Migration of internal accounting or project management records

Technology Stack

Layer	Technology	Purpos e
Design	Figma	Design system, component library, responsive layouts
Markup & styling	HTML5, CSS3, Tailwind CSS	Semantic structure, Grid/Flexbox, design tokens as config
Client scripting	JavaScript (ES2022)	Filtering, lightbox, form feedback, lazy loading — as enhancement
Server & database	Node.js, Express, MySQL	Routing, templating, sessions, normalised relational storage
Version control	Git, GitHub	Branch-per-feature workflow, review history, issue tracking
Hosting & analytics	Managed Node hosting, TLS, GA4, Search Console	Deployment, certificates, traffic and conversion measurement

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System Architecture

Client Tier — Browser

Responsive HTML · compiled Tailwind stylesheet · progressive JavaScript · lazy-loaded imagery

↓ HTTPS request / rendered HTML response ↓

Application Tier — Node.js & Express

Public routes · auth middleware · enquiry pipeline (validate → persist → notify) · authenticated admin routes

↓ parameterised queries ↓

Data Tier — MySQL

services · projects · project_images · enquiries · users — with scheduled backup

↓ served directly with cache headers ↓

Asset Layer

Processed photography in multiple widths and modern formats · fonts · compiled CSS · client scripts

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Methodology — Four Phases

An incremental, phase-wise method: each phase ended in something the firm could see and use, and its review with the Principal Architect set the priorities of the next.

Phase	Period	Focus	Deliverable at Review
1	Jan – Feb 2026	Discovery, information architecture, design system	Approved Figma system and page designs at three breakpoints
2	Feb – Mar 2026	Front-end implementation, responsive and accessible	Working static front end reviewed on real devices
3	Apr – May 2026	Express app, MySQL schema, enquiry pipeline	Database-driven site with recorded, notified enquiries
4	May – Jul 2026	Admin panel, optimisation, deployment, handover	Live platform, staff training and written documentation

PHASE 1 · JAN – FEB 2026

Discovery & Design System

- Discovery interviews with the Principal Architect and site team found that clients browse by resemblance, not by service — shaping a project archive filterable by type, not just by vertical
- Map Approval enquiries are procedural, not aesthetic — that page was designed around checklists and timelines instead of a portfolio
- Almost all traffic would arrive on mobile from Instagram, so every layout was designed at 360px first and expanded upward
- The Figma design system used a near-monochrome palette, one typographic pairing and a twelve-column grid, with every colour pair checked against WCAG 2.1 AA at definition

PHASE 2 · FEB – MAR 2026

Front-End Implementation

- Figma tokens were transferred into the Tailwind configuration first, so no spacing, type size or colour could enter the codebase outside the system
- JavaScript was written as an enhancement: every page stays readable and every form stays submittable with scripting disabled
- Testing on real devices found the enquiry call-to-action sitting outside comfortable thumb reach — layouts were revised, and every interactive target audited to 44px minimum
- Keyboard-only testing found the lightbox trapped focus incorrectly; fixing focus containment and restoration took longer than building the lightbox

PHASE 3 · APR – MAY 2026

Server, Database & Enquiry Pipeline

- The MySQL schema was normalised to third normal form and built through version-controlled migration scripts, rebuildable from scratch
- Express routes stayed thin — a route resolves parameters, calls a query function, renders a template; no SQL lives inside a route handler
- Every enquiry field is validated again on the server; a valid submission is written to the database inside a transaction before triggering an email notification, so a mail failure never loses a lead
- Rate limiting and a hidden honeypot field handle spam without a challenge; the firm could, for the first time, answer how many enquiries it received in a week and for which service

PHASE 4 · MAY – JUL 2026

Admin Panel, Optimisation & Handover

- Staff can add and edit projects, upload and reorder photographs, and review enquiries — tested by watching a staff member add a real project unaided
- Uploaded photography is processed into multiple widths in a modern format, lazily loaded, with dimensions declared to eliminate layout shift
- Every page received purpose-written metadata, a coherent heading hierarchy, local-business structured data and a database-generated sitemap
- The internship closed with production deployment, a documented backup and restore procedure, staff training, and written operating documentation

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Outcomes Achieved

A live, database-driven platform	**Achieved**
A documented, drift-free design system	**Achieved**
A structured enquiry pipeline	**Achieved**
Independent content maintenance by staff	**Achieved**
Measurable mobile performance improvement	**Achieved**
Technical search foundation	**Foundation in place, ranking accrues over time**
WCAG 2.1 Level AA accessibility	**Substantially achieved**

PROJECT DESCRIPTION

Skills Acquired

Product & UX
Discovery interviewing, information architecture, mobile-first layout

Visual Design
Design systems, type and spacing scales, component libraries in Figma

Front End
Semantic HTML5, CSS Grid/Flexbox, Tailwind, progressive enhancement

Back End
Node.js and Express, routing, middleware, server-side templating, sessions

Data
Relational modelling, normalisation, parameterised queries, migrations

Security
Input validation, injection/XSS defence, password hashing, security headers

Performance & SEO
Responsive images, caching, structured data, analytics and conversion events

Professional Practice
Client communication, scope negotiation, technical writing for non-technical readers

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Challenges and Limitations

- Working to a design-led client: feedback arrived as judgement (“a layout felt crowded”) rather than specification — converting that into a spacing token took most of Phase 1
- Photography varied widely in resolution and quality; since reshooting was not possible, the pipeline and layouts were designed to tolerate inconsistent crops
- Content was the bottleneck: writing project descriptions competed with senior staff time on live projects, so the archive launched with fewer projects than designed to hold
- Working alone across design, front end, back end and deployment meant no second technical opinion — mitigated with a checklist and self-review before every merge
- Full accessibility conformance and search ranking both depend on effort that continues after handover, not on the code delivered within the internship

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Future Scope

- **A client project portal** — per-client login showing stage of work, approved drawings and payment milestones; the existing schema and authentication already support most of it
- **A Map Approval process tracker** — a public tracker showing submission stage and required documents
- **Enquiry analytics** — a dashboard reading the data already captured: volume by service, by month, and time to first response
- **Editorial content and a structured photography standard** — case studies for local search, and a shooting brief to fix inconsistency at the source

PROJECT DESCRIPTION

Conclusion

The six-month internship at Limitless Spacez delivered a live, database-driven web platform, the design system it is built on, the enquiry pipeline that captures its leads, the administration panel through which its staff maintain it, and the documentation that lets all of this continue without me.

A platform that cannot be maintained by the people who own it will decay regardless of how well it is engineered — which is why the administration panel and the handover documentation are the parts of this project I would defend first.

This was the first occasion my work was used by people who had not asked how it was built and did not need to care. Applying that standard is the most useful thing I take from these six months into the remainder of my degree.

Thank You

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Full-Stack Developer & Design Intern, Limitless Spacez